

Implementation Spec (for Claude Code)

This tab is the implementation spec: build to these instructions directly. It does not include alternatives or rationale — see the "Design Context" tab in this doc for the reasoning behind each decision if needed.

1. Data

- Use the antibody-antigen subset of SKEMPI 2.0.
- Deduplicate homologous entries that share the same point mutation, matched by aligned interface/structural position (not raw residue number). Keep homologous entries with different mutations.
- Build two train/val splits:
 1. Held-out-PDB split — no PDB/complex in both train and val.
 2. Same-PDB-allowed split — same PDBs may appear in both, held-out mutations only.
- Report all downstream metrics separately per split. Do not pool them.
- Flag and separately track three label types per entry: bounded $\Delta\Delta G$ value, "ineq" (one-sided bound), "n.b." (no binding detected).

2. Pre-processing

- For every entry, run wild-type and mutant chains (heavy chain, light chain, antigen — as separate chains, not concatenated) through:
 - A structure model (ESMFold or IgFold) → per-residue structural embeddings.
 - A sequence model (ESM-2 or AbLang2) → per-residue sequence embeddings.
- Wild-type structure: use the real PDB structure directly. Do not compute a structural confidence score for the wild-type branch (it's not a prediction).
- Mutant structure: full model rerun on the mutated sequence (not embedding injection into the wild-type representation). Compute per-residue structural confidence (pLDDT or prmsd) for the mutant branch only.
- Compute per-residue sequence-naturalness confidence (pseudo-log-likelihood via masked-residue scoring) for both wild-type and mutant branches.

- Do not crop input to a fixed window. Keep full sequence length for both branches; instead, compute a per-residue mask/weight (using the SKEMPI support/core/rim/surface/interior interface classification) to be applied during pooling, not before.
- All of the above is precomputed once per entry, offline, before training.

3. Model

- Two-branch architecture: wild-type branch and mutant branch, each producing a per-residue representation combining structure and sequence embeddings.
- Encode the mutation explicitly as a query: 3D location, sequence position (with chain identity), and a one-hot wild-type-to-mutant residue encoding.
- Two distinct weighting mechanisms, applied at two different stages — do not conflate them:
 1. Confidence weighting (applied before subtraction). Weight each branch's per-residue representation by its own confidence: mutant branch by structural confidence times sequence confidence; wild-type branch by sequence confidence only (no structural confidence — see Pre-processing).
 2. Mutation-window masking (applied after subtraction, during pooling only). After subtracting the confidence-weighted wild-type representation from the confidence-weighted mutant representation, apply the interface-based per-residue window/mask (from Pre-processing) as the pooling weight when reducing the difference representation to a single fixed-size vector — so only residues around the mutation in the wild-type structure contribute to the pooled output. Use a smooth (e.g. Gaussian) falloff centered on the mutation site rather than a hard zero/one cutoff, so weight decays gradually with distance instead of dropping sharply at an arbitrary window edge. This masking step is not a confidence score; it is a structural relevance weight for pooling.
- Subtract the confidence-weighted wild-type representation from the confidence-weighted mutant representation to get a difference representation.
- Feed the pooled vector into the prediction head(s) (see Loss).

4. Loss

Implement a per-sample, two-term loss, selected by label type:

Implement in two staged iterations — do not implement both terms at once:

- **Iteration 1.** Train and validate using only bounded- $\Delta\Delta G$ entries. Mask out (exclude from loss and from batches) all "ineq" and "n.b." entries for this iteration — same dataset/splits, just filtered. Get this fully working and validated before starting iteration 2.
- **Iteration 2.** Once iteration 1 is validated, add the hinge loss term and include "ineq"/"n.b." entries in training, per the two-term loss below.
- Bounded entries (real $\Delta\Delta G$ value): [PLACEHOLDER — regression or bin-distance-weighted classification, to be finalized before implementation; see Design Context tab, section 4]. Apply a no-penalty margin (regression) or near-zero cost between adjacent bins (classification) sized to the empirical noise floor: 84% of independently-remeasured same-mutation pairs agree within 1 kcal/mol.
- "ineq" and "n.b." entries: hinge loss. Zero penalty when the prediction is on the correct side of the known bound; penalty grows linearly (or as specified at implementation) beyond it. Anchor "n.b." entries' threshold further out than ordinary "ineq" bounds, reflecting that binding is effectively abolished.
- Combine terms via batch-averaged sum. Apply category reweighting if bounded vs. ineq/n.b. sample counts are highly imbalanced within a batch.

4b. Staged Training & Ablation Plan

Additional staging, on top of the iteration-1/iteration-2 loss staging above:

- **Sanity check first.** Before real training, run an overfitting test on a small subset (or the full training set with no regularization) to confirm the network can actually fit the data at all. Do not proceed to real training until this passes.
- **Embedding ablation, staged.** Pre-compute structure and sequence embeddings separately and independently (already required — see Pre-processing). At the modeling level, support selecting which embedding source(s) feed the model at training time. Start real training using structure embeddings only; in a later cycle, add sequence embeddings back in and compare. This isolates whether/how much the sequence modality actually helps, rather than assuming it from the start.

5. Metrics

Report all metrics separately for each of the two splits (held-out-PDB, same-PDB-allowed). Do not pool across splits.

- Bounded-label head: MAE/RMSE and Spearman correlation (regression) or accuracy and adjacency-aware confusion matrix (classification), reported alongside the 84%-within-1-kcal/mol noise-floor figure as a reference ceiling.
- Inequality/hinge head: correct-side accuracy, computed separately for "ineq" and "n.b." entries.
- Run and report two additional analyses: (1) mutation location (support/core/rim/interior/surface) vs. $\Delta\Delta G$; (2) $\Delta\Delta G$ distribution comparison, alanine-scanning (X $\rightarrow$ A) mutations vs. other single-point substitutions.